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Museum accessibility: a managerial perspective on digital approach

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***Abstract*— Accessibility is one of the cornerstones of a modern museum enhancement, a feature highlighted by many policymakers, which also translates as a greater openness to different museum audiences. Among these public, the one represented by vulnerable people is linked to a precise management, based mainly on the removal and overcoming of physical and sensory barriers and on the use of the experience related to the works. In the museum field, the pandemic has accelerated the processes of digitization, leading many companies to develop strategies and practices to make their heritage accessible also at the digital level. This work emphasizes that this greater accessibility to museums linked to digital, is not limited only to a process of reproduction of works or places but increases the experience of the visit with additional information that is integrated with the basic experience, enabling the user to active value co-creation processes regarding the visit experience. A qualitative data analysis was carried out by using a multiple case study, analyzing top museums to understand which are the strategies to improve accessibility to the most fragile public and what impact these strategies can have on management: to do this, a logic of repetition, in order to highlight the differences and similarities. These findings contribute to a better understanding of accessibility in cultural institutions and a series of managerial implications are proposed.**

Keywords—museum, accessibility, digital, virtual tour, vulnerable audience, smart service

I. Introduction

Fair and inclusive accessibility to cultural heritage is one of the most important themes of contemporary social and cultural debate. It is an awareness enucleated by the value of culture, the role that the arts occupy for the development of identity and collective territories of which every single social actor can and must benefit. The intrinsic link between the cultural value and the commitment of the social actors, be they individual or community, is recognized and confirmed by national and international political decision-makers, first of all the European Union, which already in the Faro Convention [1] Declares that "everyone, alone or collectively, has the right to benefit from cultural heritage and to contribute to its enrichment", and that they are committed to "fostering an economic and social climate that supports participation in cultural heritage activities"; "The Parties undertake to: promote actions to improve access to cultural heritage, in particular for young people and disadvantaged people, in order to increase awareness of its value, the need to preserve and preserve it and the benefits it can bring derive". As reported in the European Cultural Heritage Action Framework, culture is a driving force for the development and consolidation of the sense of belonging to a community "based on common cultural heritage, historical experiences and shared values - to make the most of this potential, we must ensure that everyone has access to the widest possible cultural heritage, in all its forms" [2]. Starting from supranational and national directives, numerous studies and experiments are contributing to scientific progress in the field of cultural participation accessible to every single social actor, including those disadvantaged or hampered by physical, sensory or cognitive barriers that reduce or impede access to cultural heritage [3]. This work aims to contribute to the theme through an exploratory study on the adoption by museum management of accessibility strategies aimed at the inclusion of the public. The study therefore aims to highlight how the accessibility of cultural offerings, facilitated by digital transformation, enables people to break down barriers resulting from different types of disabilities, add meaning to works of art, and extend and deepen their understanding of cultural heritage.

II. Literature review

A. *The role of culture in social inclusion and community wellbeing*

In a global scenario in which the international socio-economic paradigms are subject to radical changes, culture, by reason of its intrinsic, evolutionary and transformative nature, is one of the major activators and drivers of economic, social and technological progress of a country. The role of culture within the complex socio-economic ecosystem in which it is located is the subject of recent attention by policy makers as well as scholars and practitioners. Specifically, it is from the discourse on cultural policy that organizational and social systems emphasize the profound interconnection between access to cultural enjoyment and quality of life [4]. These are complex two-way processes from whose dialogue emerges the empirical awareness that culture plays a decisive role in improving the quality of life and local well-being. This is confirmed by the therapeutic declination recently opened by cultural policy [5]. In this direction, the concept of cultural heritage and consequently its role within society is also subject to a paradigm shift. The classical vision of culture based on the exclusive conservation and exhibition of works of art is widening its boundaries to a participatory and inclusive space [6] in which culture is enjoyed, produced and co-produced by each stakeholder involved in the process of creating cultural value. The ongoing paradigm shift is not only reflected in the internal transformation of the mechanisms of conservation, use and enhancement of art collections but also of the subjects

that in various ways gravitate in the orbit of cultural and creative organizations. The complexity of cultural management is inextricably linked to the heterogeneity of the needs, expectations and interests of each individual stakeholder [7]. All those social actors who are disadvantaged in their ability to co-create value and, consequently, intensify their well-being [8][9] are part of the scenario outlined. In this trajectory it should be noted that despite the openness of cultural organizations to the reception of needs and expectations coming from heterogeneous targets with each other, there is, to date, a lack of attention on the part of the scientific community towards cultural participation for sections of the population affected by disadvantaged audiences, and in particular those with disabilities [10] physical, sensory, cognitive. The attention of the cultural and creative sector to the holistic inclusion of this particular group of disadvantaged audiences highlights the opening of a deep gap in the consideration of the needs of this target [11] making it in most cases invisible to the attention of cultural management. An exception to this are high-level cultural and artistic organizations that benefit from the size, attractiveness and recognition of significant state support [12]. Attention, however, registers critical issues of considerable importance. In particular, the activities and initiatives implemented by cultural and creative organisations in order to include disadvantaged sections of the population in cultural enjoyment by ensuring access to it, denounces a lack of equity in the representation of citizens with disabilities compared to able-bodied citizens [13]. In this regard, it should be noted that the accessibility of cultural knowledge derives from an intrinsic link between the individual characteristics of the subjects who benefit from culture and the environment in which they are inserted [14]. Understanding the role that culture plays in the lives of people with disabilities is also crucial for disability studies [15][16] and the strategic and managerial trajectories to be undertaken to facilitate each individual category that makes up this segment of the population in social and cultural participation [17]. Redefining the access modalities in an inclusive way, identifying and experimenting innovative models of managing culture, enhances the contribution it makes to the well-being of citizens and the development of the territories in which cultural value is conveyed. In fact, to benefit from the cultural value conveyed by cultural and creative organizations that choose to adopt inclusive and accessible approaches to artistic enjoyment are not only visitors with disabilities but every single visitor [18][19][20]. It is now clear that an accessible participation in the cultural heritage is able to facilitate a positive sense of identity in people with disabilities because of the promotion of diversity and the transformation of the key to reading [21]. Social inclusion, on the other hand, constitutes a significant dimension of well-being and of perceived health [22]. Curbing exclusion through effective and efficient management of cultural participation practices is, therefore, a fundamental pillar for full social inclusion [23] in which every single social actor has the opportunity to develop a solid sense of community [24].

B. Technology for accessible cultural participation

Accessibility is a subject of significant cross-sectoral interest. The cultural and creative sector is not without the attention dedicated to the opportunity to develop strategic trajectories of participation and interaction with people with disabilities [25] or who have seen their skills decrease. However, the focus on the topic is still in its early stages in terms of specifically targeted research strands delineated by physical disabilities, sensory and cognitive that hinder access to places of culture and consequently prevent the fruition [26]. A decisive contribution to scientific advances in cultural accessibility is the use of new technologies in a widespread, customizable and shared way [27]. The cultural and creative sector, and in particular the museums, are particularly attentive to the potential deriving from the use of digital innovations [28], especially useful for extrapolating specific information on one's audience and therefore improving a customized offer [29]. The managerial literature returns in fact relevant perspectives in relation to the dynamics susceptible to verification by the dialogue between culture and digital [30][31]. Focusing the attention on the accessible declination of cultural value enabled by digital technologies, it is necessary to note the changes implemented by cultural and creative organizations, and in particular by museums, to ensure the full participation of people with disabilities in cultural enjoyment [32]. In this direction, the role played by the narrative that takes place inside and outside the boundaries of the museum so that accessibility, inclusion and participation constitute the faces of the same prism [33]. From a managerial point of view, it is crucial to draw up strategic and operational plans aimed at diversifying the cultural offer. Diversification aims to respond to different types of accessibility, physical, functional, perceptive-sensory, to ensure equity of participation and involvement to each individual stakeholder involved in the processes of creation and co-creation of cultural value. In the case of museum institutions, the use of enabling technologies allows management to adopt an inclusive approach to cultural participation, involving groups of the population hindered to enjoy cultural value due to physical barriers, sensory, cognitive. It is widely believed in the scientific community that digital technologies are powerful growth tools that can reach and involve a wide and heterogeneous audience for interests and preferences, which translates, among other things, in expanding access to cultural activities [34]. In this direction, the role played by digital technologies in promoting accessible cultural participation is decisive. Firstly, technology allows you to get in touch with a large amount of information available free of charge and within seconds, which in the case of accessible tourism has significant value [35]. Moreover, digital technologies constitute the structure of interactive and sensory environments through which to adapt the physical space of a museum to the user with disabilities [36]. Finally, devices such as frameworks, APIs or repositories have been developed, thanks to which it is possible to enjoy fully accessible mobile applications [37]. Mobile devices, in particular, offer huge potential for the promotion of accessibility especially for people with visual disabilities, as they incorporate a number of accessories, sensors or instruments that can efficiently meet the needs of this category [38]. However, there is little scientific input on the role of assistive technologies in the museum sector [39]. In the case of hearing disabilities, involving deaf and hearing impaired visitors, communication support technologies have been developed that can process real-time subtitles via speech recognition [40]. The heterogeneity of the technology driven tools and mechanisms tested is such as to respond to different accessibility needs. Among the facilitators that today are an integral part of the exhibition spaces of museums and archaeological sites [41] should be considered recorded guides, touch tours, manipulation sessions, tactile plans, information on large print and Braille, tours interpreted in sign

language [42]. If effectively assisted by digital and cognitive technologies, the initiatives and practices aimed at promoting an accessible cultural participation is able to foster a development of identity as well as the well-being of the actors involved [43]. Therefore, if there are different ways to interact with the visitor even in a non-digital way, such as the interpretive materials that accompany visitors in the experience, such as labels, panels, gallery cards, signs [44], digital brings the advantage of being considered easier to update or change, both economically and in terms of sustainability [45]. Technologies such as 3D printers can have an important impact in terms of accessibility for blind visitors, not only by reproducing the works, but also by modifying them in scale, in order to make very large works usable [46]. But he objective of these technologies cannot be just the mere use, but the complete transformation of the experience for this type of public through interaction [47]. This is a change that technologies must aim for, transforming from facilitators of accessibility into tools of inclusion [48].

III. Methodology

The exploratory purpose of this analysis is consistent with the use of a qualitative research methodology [49]. Through the method of multiple case studies [50] the authors aim to highlight the presence or absence of good practices, aimed at fostering accessibility through the implementation of innovative technologies, carried out by major international museums, in the perspective of a future in-depth study on the type of services offered. The 10 major museums identified by Museum World Ranking website and listed below were chosen as the object of analysis: Louvre Museum (Paris), National Museum of China (Beijing), British Museum (London), National Gallery (London), Metropolitan Museum of Art (New York), Vatican Museums (Rome), Forbidden City (Beijing), Tate Modern (London), Natural History Museum (London) and State Hermitage Museum (St. Petersburg). The aforementioned website ranks museum institutions using an exhaustive methodology based on criteria of effectiveness, managerial and organizational culture, profitability, competitiveness, innovation, competence, prestige and internationalization. In particular, the decision to rely on this classification lies in the fact that it was developed on the basis of quantitative research conducted through questionnaires concerning, among others, accessibility by different target groups of visitors, including the senior citizens, children and persons with disabilities. A subdivision of the services that museums offer to vulnerable audience was made, based on the categories that emerged in the literature: physical, sensory (auditory and visual) and cognitive (developmental and learning disabilities) disabilities. A further classification has been made where among these services, those that use digital technologies are highlighted in order to better highlight these tools.

IV. Findings

The analysis of the museums observed has highlighted a series of actions that museums have undertaken in offering services to the most vulnerable audience (Tab.1). It was therefore highlighted how among these services, they have a use of technology and gender. The Louvre Museum in Paris, for the physically disabled, offers a dedicated assistance service, with the possibility of following an established route with specific elevators. For those with hearing disabilities, there is no possibility of using hearing loop systems, available at the entrance for the hearing impaired. There is also the possibility of requesting an audio guide on Nintendo 3DS XL with audio description, which illustrates the most famous works, as well as giving the possibility of using the map in an interactive way, letting the user choose what he prefers to know. As regards visitors with visual impairments, the museum offers two dedicated spaces, with the use of Hands-on devices dedicated to the evolution of the museum over the years and tactile devices for the Islamic Art Collection. With regard to other types of disabilities, the museum provides dedicated accompanying services. There are also rooms and spaces with specific lights, as well as the possibility of having a SensoryBags, with kits containing material for people with sensory impairments: noise-cancelling headphones, eye bands, etc. The National Museum of China provides few services for fragile audiences. In addition to providing wheelchair assistance, the museum offers smart guides for advice on the inclusiveness of the museum, such as the best routes to choose. The British Museum in London offers various services dedicated to the various types of disabilities, also taking advantage of the AccessAble platform, where the services are performed. The museum also offers the possibility of downloading activities to be carried out at home for children unable to physically come to the museum. For hearing-impaired visitors, British Sign Language Guides are available on the British Museum app. Visually-impaired visitors can instead take advantage of Audio Descriptive Guides on their device. There are also some "Hands on desks" moments where the museum volunteers allow you to touch some objects from 11.00 to 16.00. You can also book a touch tour. For other types of disability, sensory maps are available online that indicate the differences in lights, smells or where the noisiest areas are. Sensory support backpacks are available upon request to support young visitors with additional sensory needs. SEN (Special Educational Needs) students use digital technology and multisensory activities to explore the collection thanks to dedicated workshops, which are organized periodically. The National Gallery in London provides a map on its website with all the references for people with physical disabilities. For people with hearing impairments, there are hearing loops for talks and films in the Sunley Room and exhibition cinemas. British Sign Language interpreted talks and Live speech-to-text captioning. For visually impaired visitors, the exhibition videos and films have subtitles and transcripts with large print transcripts. The Metropolitan Museum of Art in New York provides a number of support services. Visitors with hearing problems have hearing loops, free audio guides and some talks on specific topics, accompanied by sign language. They have specific services for the visit, but American Sign Language interpretation is available for virtual Museum events and online resources with two weeks' notice. For visually impaired visitors, specific physical and virtual tours are available for the blind, as well as some dedicated workshops with the possibility of touching specific objects. For visitors with other types of disabilities, there is a section of the site with advice for the visit, maps (including a Sensory Friendly Map) and photographic references on the route to facilitate the visit. The Vatican Museums provide visitors with motor disabilities with a map for a barrier-free itinerary. Deaf or partially deaf visitors can always take advantage of a Video Guide in American Sign Language. On some days and time slots, three types of free tours are available, guided in Italian Sign Language. For blind or partially blind visitors, a tactile and multi-sensory tour of some areas of the museums is available free of charge. In Beijing's Forbidden City, disabled people are provided with a

wheelchair route map. There is also the possibility of visiting the site virtually with the Digital Palace Museum. The Tate Modern of London people with motor disabilities have the possibility to park and request scooters to move around the museum (notifying 24 hours in advance). The site indicates all types of support for visits by disabled people (bathrooms, changing places toilets, lifts, points for resting). For the hearing impaired, the ticket desks are fitted with a hearing loop. There are online guides in larger print for people with limited vision and magnifiers. For blind users, description tours and descriptive video material are available. As far as the other types of disabilities are concerned, a specific visual guide to the museum structure is available, with visual information, particularly suitable for people with autism. The museum also provides a quite room for neurodiverse people, families or anyone who would like to spend some quiet time away from the gallery environment. Communication cards (to print or in digital format) are also available on the website in order to provide a means of visual, or nonverbal, communication about directions, toilets, coffee and other information. The Natural History Museum of London provides disabled facilities such as parking, wheelchairs, lifts and toilets. For people with hearing impairments, learning spaces and audio visual displays are equipped with an induction loop system. In addition there are regular British Sign Language (BSL) signed events at the Museum. Tools are available to facilitate visits for people with visual impairments such as large-print versions of the Museum map, event schedule and selected exhibition guides. The State Hermitage Museum in St. Petersburg does not provide any type of tools for people with disabilities, except for a map with the routes reserved for people with motor disabilities.

TABLE I. SERVICES FOR THE DISABLED

Musem	*Physical Disabilities*	*Sensorial Disabilities*	*Cognitive Disabilities*
Louvre	-Assistance and specific paths	-Hearing loop -Nintendo 3DS XL -Hands-on devices and tactile devices	-Dedicated accompaniment -Dedicated lights -Sensory Bags
National Museum of China	-Wheelchair assistance	N/A	N/A
British Museum	-Dedicated assistance -Remotely activities for children	- App with BSL guide -Audio Descriptive Guides -Hands on desks -Touch tour	-Maps -Sensory support -Digital workshop for SEN
National Gallery	-Assistance and specific paths	-Hearing loops -Talks in BSL -Live speech-to-text captioning -Subtitles and transcripts for videos -Large print transcripts.	N/A
Metropolitan Museum of Art	-Assistance and specific paths	-Sign Language for virtual events -Hearing loops -Audio guide -Talk accompanied by sign language -Specific physical and virtual tours -Dedicated workshops	-Tips for visits -Various maps
Vatican Museums	-Barrier-free map	-Video Guide in Sign Language -Tour in Sign Language -Tactile and multi-sensory tour	N/A
Forbidden City	-Digital Palace Museum -Wheelchair route map	N/A	N/A
Tate Modern	-Various and informative assistance	-Hearing loop -Online guides in larger fonts -Magnifiers -Description tours -Descriptive videos	-Dedicated online guide -Quite room -Communication cards
Natural History Museum	-Various assistance and support services for the visit	-Loop system -Sign Language for the events -Large-print versions of all the material	N/A
State Hermitage Museum	-Map with the paths reserved for the disabled	N/A	N/A

Source: author's elaboration

Although not created for an audience of fragile people with disabilities, these museums present some digital services that are particularly useful especially for this type of audience. In fact, all the museums analyzed have the possibility of displaying part of their digitized collections, with insights into the individual works. The public with physical disabilities can in fact take advantage of virtual tours in most cases. The Louvre Museum offers its visitors the possibility of using a Nintendo 3DS XL as a guide, which thanks to its screen is particularly useful for deaf visitors thanks to texts and videos. Tours of different sections are also available on the site, as well as activities from home, such as podcasts, online events, a section for kids with explanations and a VR on the Mona Lisa. The National Museum of China has several virtual tours available on its website. Smart guides are also available which also provide advice on inclusiveness within the museum. The British Museum does not currently provide virtual tours, but presents an app and interactive guides designed for visually impaired and hearing impaired audiences. The museum also offers Digital Workshops specially designed for SEN students. The National Gallery provides various types of themed virtual tours and the ability to wander freely with Google street view. The Metropolitan Museum of Art offers The Met 360° Project, with the ability to navigate different 360° environments. MetKids is a project made for, with and by kids which includes maps, dedicated videos and an interactive educational section. Virtual tours for blind and Sign Language during the virtual events are also available. The Vatican Museum offers on its website the possibility of a 360° Virtual tour of some of the most important environments, as well as a video Guide in Sign Language. The Forbidden City offers Digital Palace Museum with some of the digitized museum environments of the palace, as well as thematic digital insights. At the Tate Modern, although there is a video of a 360° environment, accompanied by some explanations, it appears to be an isolated project. The site presents Online guides with enlarged characters, for easier reading and dedicated videos. The Natural History Museum in London offers a 360° virtual tour on Google Arts & Culture with some itineraries. The State Hermitage Museum presents the Virtual Visit on its website where most of the rooms have been

digitized with information for users, and it is possible to navigate freely.

TABLE II. SERVICES FOR THE DISABLED

Musem	*Physical Disabilities*
Louvre	-Nintendo 3DS XL -Virtual tours -VR on the Mona Lisa
National Museum of China	-Virtual tours -Smart guide
British Museum	-Museum app -Digital Workshop
National Gallery	-Virtual Tours
Metropolitan Museum of Art	-The Met 360° Project -MetKids -Virtual tours for blind
Vatican Museums	-360° Virtual tour -Video Guide
Forbidden City	-Digital Palace Museum
Tate Modern	-Online guides -Dedicated video
Natural History Museum	-360° virtual tour
State Hermitage Museum	-Virtual Visit

Source: author's elaboration

While it is clear that most museums have developed virtual tours, these tours, as previously mentioned, are not developed exclusively for users with disabilities, but are considered a very useful tool for those with physical disabilities. The virtual tour cannot be considered a substitute tool for the physical experience [51], but it can be considered a particularly effective tool for those who have difficulty physically traveling on site. It is also interesting to note how many of these subjects have developed digital tools for people with visual or hearing problems. Among them many of these tools can be used on museum sites, while some are available for physical visit. It is also interesting to note that a significant number of these museums dedicate tools for this type of public not for visits, but for events in the museums (Tab.2). There is therefore a particular difference between the Western approach and that of other countries (China and Russia): in the first case, numerous types of disability are treated, using a digital approach, as seen, in some cases; in the second case, digital and non-digital services focus exclusively on physical disabilities, not offering support to other types.

V. IMPLICATIONS

This study aims to improve the knowledge and awareness of the different services that can be implemented to improve the museum visit experience of vulnerable people. Thanks to the analysis of the cases, it has been possible to draw a framework of the facilities implemented for vulnerable people, making a distinction between Western museums, which are more attentive to the development of services dedicated to visitors with different types of disabilities, and Eastern institutions, with limited facilities. From a theoretical point of view, the implementation of accessible tourism, which is not a secondary objective for major international attractors such as those analyzed, can have a greater chance of success when it is also of benefit to the tourist in general and integrated into the main offerings. Many of the digital services identified are able to enrich the experience of all types of visitors, increasing the quality of the museum offer. From a managerial point of view, this contribution highlights how innovative technologies can be considered an important driver to ensure an adequate enjoyment of cultural heritage by vulnerable actors, influencing the business strategies underlying the cultural offer [52]. A key task for cultural decision-makers is to encourage investment in accessibility and to demonstrate its economic and social benefits to the whole community. It is also crucial that all staff members acquire a solid knowledge base on accessibility through good knowledge management. This can be supplemented with information on good practice examples and with the experience of external experts in order to learn from the experience of other successful providers. Regular training of staff and management is also important to keep all service providers up-to-date and to help them understand the demands and wishes of visitors. Many service providers still feel uncertain about how to treat a disabled guest or about the specific needs of children and elderly people and the type of technology to use. This is why many guests experience attitudinal barriers and find the way they are treated an important aspect of their journey. Special training involving guests from different groups is very useful and can ensure a continuous exchange between guests and providers, ultimately leading to a better quality of service.

VI. FINAL REMARKS AND FUTURE RESEARCH

This study contributes to the current academic debate on museum accessibility, by providing a description of services for vulnerable people, and highlights how the accessibility of cultural offerings, facilitated by digital transformation, enables people to break down barriers resulting from different types of disabilities, add meaning to works of art, and extend and deepen their understanding of cultural heritage. Knowing the impact that new technologies have on the management of vulnerability is crucial for any cultural manager and in recent years accessible tourism practices have multiplied, providing services that can be used by both vulnerable actors and non. The exploratory analysis presented in this study will need to be deepened with furtherture research that can identify and classify the different cultural services offered and broaden the scope of investigation to a larger sample of museum institutions. These insights can provide cultural managers with an integrated vision and the fundamental elements for the creation of an environment for service quality.

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